
```

% Sweep Newton initial guess from mu to 90 deg
mu = asin(1 / M);
theta0Grid = linspace(mu, pi / 2, 200);

weakRoot = weakNewton;
strongRoot = strongNewton;
rootTol = 1e-4;

outcome = strings(numel(theta0Grid), 1);
finalRoot = nan(numel(theta0Grid), 1);
funcCountFinal = nan(numel(theta0Grid), 1);

for i = 1:numel(theta0Grid)
    theta0 = theta0Grid(i);

    try
        [xNR, infoNR] = newton_raphson(f, fprime, [theta0 theta0], atol,
maxit);
        finalRoot(i) = xNR;
        funcCountFinal(i) = infoNR.history.funcCount(end);

        if abs(xNR - weakRoot) <= rootTol
            outcome(i) = "Weak";
        elseif abs(xNR - strongRoot) <= rootTol
            outcome(i) = "Strong";
        else
            outcome(i) = "Fail";
        end
    catch
        outcome(i) = "Fail";
    end
end

T = table(rad2deg(theta0Grid(:)), outcome, finalRoot, funcCountFinal, ...
    'VariableNames', {'theta0_deg', 'Outcome', 'FinalRoot_rad',
'FuncCount'});
disp(T)

% Plot outcome vs initial guess
figure
hold on

idxFail = outcome == "Fail";
idxWeak = outcome == "Weak";
idxStrong = outcome == "Strong";

scatter(rad2deg(theta0Grid(idxFail)), zeros(nnz(idxFail), 1), 20, 'r',
'filled', ...
    'DisplayName', 'Fail')
scatter(rad2deg(theta0Grid(idxWeak)), ones(nnz(idxWeak), 1), 20, 'g',
'filled', ...
    'DisplayName', 'Weak')

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scatter(rad2deg(theta0Grid(idxStrong)), 2 * ones(nnz(idxStrong), 1), 20,
'b', 'filled', ...
'DisplayName', 'Strong')

grid on
yticks([0 1 2])
yticklabels({'Fail', 'Weak', 'Strong'})
xlabel('Initial guess \theta_0 (deg)')
ylabel('Convergence outcome')
title('Newton-Raphson basin of attraction')
legend('Location', 'best')

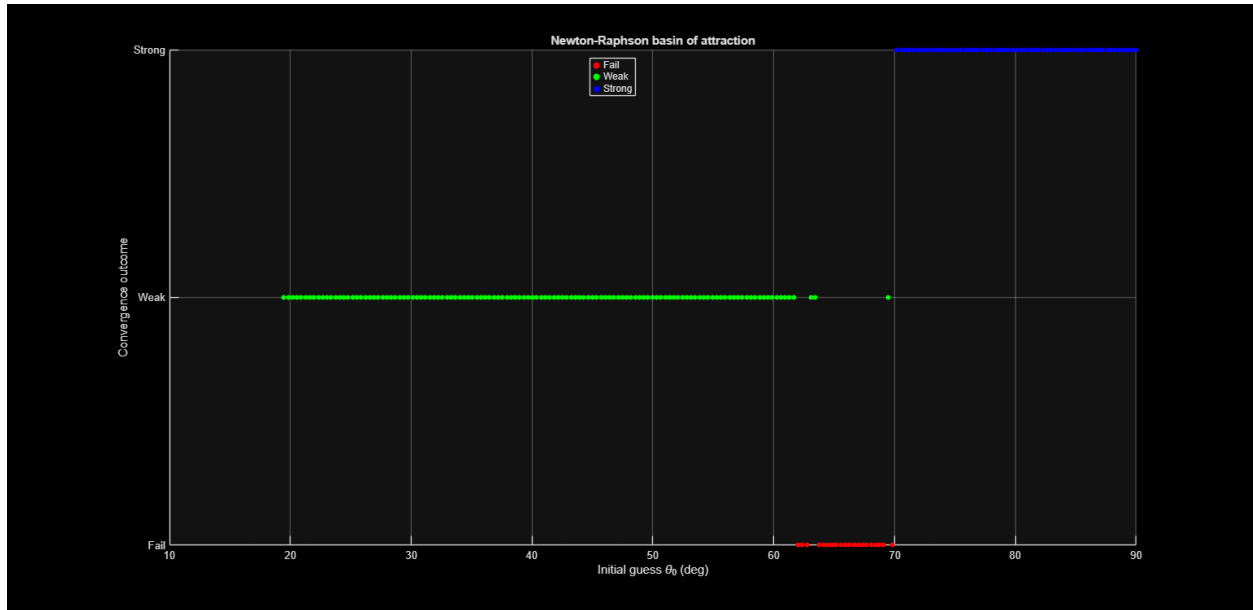
```

<i>theta0_deg</i>	<i>Outcome</i>	<i>FinalRoot_rad</i>	<i>FuncCount</i>
19.471	"Weak"	0.6591	10
19.826	"Weak"	0.6591	10
20.18	"Weak"	0.6591	10
20.534	"Weak"	0.6591	10
20.889	"Weak"	0.6591	10
21.243	"Weak"	0.6591	10
21.598	"Weak"	0.6591	10
21.952	"Weak"	0.6591	10
22.307	"Weak"	0.6591	10
22.661	"Weak"	0.6591	10
23.015	"Weak"	0.6591	10
23.37	"Weak"	0.6591	10
23.724	"Weak"	0.6591	10
24.079	"Weak"	0.6591	10
24.433	"Weak"	0.6591	10
24.787	"Weak"	0.6591	10
25.142	"Weak"	0.6591	10
25.496	"Weak"	0.6591	8
25.851	"Weak"	0.6591	8
26.205	"Weak"	0.6591	8
26.56	"Weak"	0.6591	8
26.914	"Weak"	0.6591	8
27.268	"Weak"	0.6591	8
27.623	"Weak"	0.6591	8
27.977	"Weak"	0.6591	8
28.332	"Weak"	0.6591	8
28.686	"Weak"	0.6591	8
29.04	"Weak"	0.6591	8
29.395	"Weak"	0.6591	8
29.749	"Weak"	0.6591	8
30.104	"Weak"	0.6591	8
30.458	"Weak"	0.6591	8
30.813	"Weak"	0.6591	8
31.167	"Weak"	0.6591	8
31.521	"Weak"	0.6591	8
31.876	"Weak"	0.6591	8
32.23	"Weak"	0.6591	8
32.585	"Weak"	0.6591	8
32.939	"Weak"	0.6591	8

33.293	"Weak "	0.6591	8
33.648	"Weak "	0.6591	8
34.002	"Weak "	0.6591	8
34.357	"Weak "	0.6591	8
34.711	"Weak "	0.6591	8
35.066	"Weak "	0.6591	8
35.42	"Weak "	0.6591	8
35.774	"Weak "	0.6591	8
36.129	"Weak "	0.6591	8
36.483	"Weak "	0.6591	6
36.838	"Weak "	0.6591	6
37.192	"Weak "	0.6591	6
37.546	"Weak "	0.6591	6
37.901	"Weak "	0.6591	6
38.255	"Weak "	0.6591	6
38.61	"Weak "	0.6591	6
38.964	"Weak "	0.6591	6
39.319	"Weak "	0.6591	8
39.673	"Weak "	0.6591	8
40.027	"Weak "	0.6591	8
40.382	"Weak "	0.6591	8
40.736	"Weak "	0.6591	8
41.091	"Weak "	0.6591	8
41.445	"Weak "	0.6591	8
41.799	"Weak "	0.6591	8
42.154	"Weak "	0.6591	8
42.508	"Weak "	0.6591	8
42.863	"Weak "	0.6591	8
43.217	"Weak "	0.6591	8
43.572	"Weak "	0.6591	8
43.926	"Weak "	0.6591	8
44.28	"Weak "	0.6591	8
44.635	"Weak "	0.6591	8
44.989	"Weak "	0.6591	8
45.344	"Weak "	0.6591	8
45.698	"Weak "	0.6591	8
46.052	"Weak "	0.6591	8
46.407	"Weak "	0.6591	8
46.761	"Weak "	0.6591	8
47.116	"Weak "	0.6591	8
47.47	"Weak "	0.6591	8
47.824	"Weak "	0.6591	8
48.179	"Weak "	0.6591	8
48.533	"Weak "	0.6591	8
48.888	"Weak "	0.6591	8
49.242	"Weak "	0.6591	8
49.597	"Weak "	0.6591	10
49.951	"Weak "	0.6591	10
50.305	"Weak "	0.6591	10
50.66	"Weak "	0.6591	10
51.014	"Weak "	0.6591	10
51.369	"Weak "	0.6591	10
51.723	"Weak "	0.6591	10
52.077	"Weak "	0.6591	10

52.432	"Weak "	0.6591	10
52.786	"Weak "	0.6591	10
53.141	"Weak "	0.6591	10
53.495	"Weak "	0.6591	10
53.85	"Weak "	0.6591	10
54.204	"Weak "	0.6591	10
54.558	"Weak "	0.6591	10
54.913	"Weak "	0.6591	10
55.267	"Weak "	0.6591	10
55.622	"Weak "	0.6591	10
55.976	"Weak "	0.6591	10
56.33	"Weak "	0.6591	10
56.685	"Weak "	0.6591	10
57.039	"Weak "	0.6591	10
57.394	"Weak "	0.6591	10
57.748	"Weak "	0.6591	10
58.103	"Weak "	0.6591	10
58.457	"Weak "	0.6591	12
58.811	"Weak "	0.6591	12
59.166	"Weak "	0.6591	12
59.52	"Weak "	0.6591	12
59.875	"Weak "	0.6591	12
60.229	"Weak "	0.6591	12
60.583	"Weak "	0.6591	12
60.938	"Weak "	0.6591	14
61.292	"Weak "	0.6591	16
61.647	"Weak "	0.6591	22
62.001	"Fail "	-0.17297	14
62.356	"Fail "	-0.17297	12
62.71	"Fail "	-0.17297	16
63.064	"Weak "	0.6591	18
63.419	"Weak "	0.6591	14
63.773	"Fail "	-1.7079	14
64.128	"Fail "	-2.4825	14
64.482	"Fail "	-3.3146	18
64.836	"Fail "	-8.7657	14
65.191	"Fail "	-68.456	16
65.545	"Fail "	12.393	12
65.9	"Fail "	6.1102	18
66.254	"Fail "	4.5753	12
66.609	"Fail "	3.8007	12
66.963	"Fail "	2.9686	12
67.317	"Fail "	2.9686	20
67.672	"Fail "	3.8007	16
68.026	"Fail "	3.8007	14
68.381	"Fail "	3.8007	12
68.735	"Fail "	7.7169	12
69.089	"Fail "	-1.7079	12
69.444	"Weak "	0.6591	12
69.798	"Fail "	-0.17297	18
70.153	"Strong"	1.4337	16
70.507	"Strong"	1.4337	14
70.862	"Strong"	1.4337	12
71.216	"Strong"	1.4337	12

71.57	"Strong"	1.4337	10
71.925	"Strong"	1.4337	10
72.279	"Strong"	1.4337	12
72.634	"Strong"	1.4337	12
72.988	"Strong"	1.4337	12
73.342	"Strong"	1.4337	12
73.697	"Strong"	1.4337	10
74.051	"Strong"	1.4337	10
74.406	"Strong"	1.4337	10
74.76	"Strong"	1.4337	10
75.115	"Strong"	1.4337	10
75.469	"Strong"	1.4337	10
75.823	"Strong"	1.4337	10
76.178	"Strong"	1.4337	10
76.532	"Strong"	1.4337	10
76.887	"Strong"	1.4337	10
77.241	"Strong"	1.4337	10
77.595	"Strong"	1.4337	10
77.95	"Strong"	1.4337	10
78.304	"Strong"	1.4337	10
78.659	"Strong"	1.4337	10
79.013	"Strong"	1.4337	8
79.368	"Strong"	1.4337	8
79.722	"Strong"	1.4337	8
80.076	"Strong"	1.4337	8
80.431	"Strong"	1.4337	8
80.785	"Strong"	1.4337	8
81.14	"Strong"	1.4337	8
81.494	"Strong"	1.4337	8
81.848	"Strong"	1.4337	6
82.203	"Strong"	1.4337	6
82.557	"Strong"	1.4337	6
82.912	"Strong"	1.4337	8
83.266	"Strong"	1.4337	8
83.621	"Strong"	1.4337	8
83.975	"Strong"	1.4337	8
84.329	"Strong"	1.4337	8
84.684	"Strong"	1.4337	8
85.038	"Strong"	1.4337	8
85.393	"Strong"	1.4337	8
85.747	"Strong"	1.4337	8
86.101	"Strong"	1.4337	8
86.456	"Strong"	1.4337	8
86.81	"Strong"	1.4337	8
87.165	"Strong"	1.4337	8
87.519	"Strong"	1.4337	8
87.874	"Strong"	1.4337	8
88.228	"Strong"	1.4337	10
88.582	"Strong"	1.4337	10
88.937	"Strong"	1.4337	10
89.291	"Strong"	1.4337	10
89.646	"Strong"	1.4337	10
90	"Strong"	1.4337	10



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